

n8n AI Automations

Engineered for Scale



Technical architecture dossier · v1.0

Production n8n Architecture

Three real-world enterprise pipelines built for scale, reliability and cost control.

- **SEO content engine**
Perplexity · GPT-4o-mini · WordPress
- **Instagram lead acquisition & CRM sync**
Apify · Gemini · HubSpot
- **LinkedIn outreach with AI avatar video**
Apify · ChatGPT API · Synthesia

Core stack

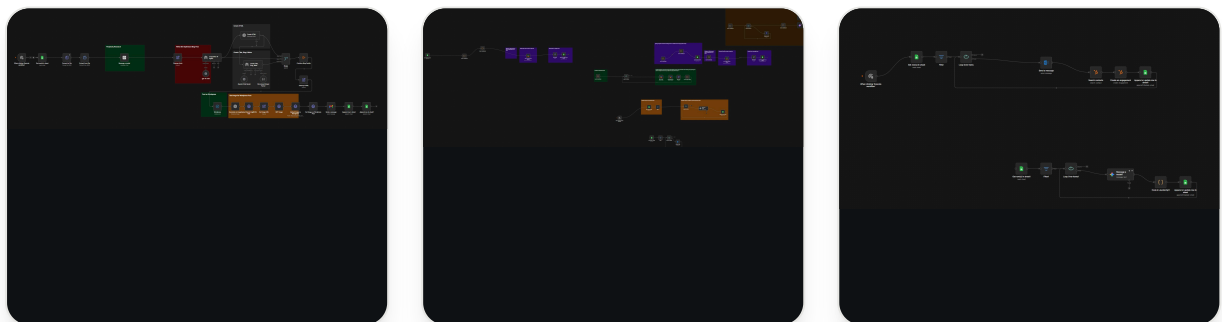


About this document

These are three automation systems running in production, not proofs of concept. Each one connects several third-party APIs, at least one LLM, and a business system of record. Each handles failure, monitors its own spend, and operates within defined compliance boundaries.

The engineering work in these pipelines is not the happy path. Wiring an LLM to an API is a tutorial exercise. The difficulty is everything around it: what happens when a scraper returns an empty payload, what a single lead actually costs before a campaign launches, and what stops a generative model from asserting something about a real person that isn't true.

Each case study follows the same structure: the business bottleneck, the architecture, the engineering decisions that made it viable, and the guardrails that make it safe to run unattended.



Production canvases, running in n8n.

Engineering principles

The three systems share a common design posture.

Cost is a first-class design constraint

Token spend and scraping volume are budgeted before execution, not reconciled after the invoice. Batching and segmentation are architectural decisions, not optimisations added later.

Failure is expected

Third-party scraping and video-render APIs have meaningful downtime. Every external call sits behind retry logic with escalation, so a transient outage does not require manual intervention.

The model does not get the last word

Generative output that touches a real person, a real company, or a published asset is constrained at the prompt level and bounded by what upstream data actually returned.

The system of record stays authoritative

Automation writes into the CRM and the operational sheet rather than beside them. An action with no recorded trace is treated as an action that did not happen.

SEO content engine

Challenge

Editorial output was constrained by a serial human bottleneck: research a topic, draft the article, format it for the CMS, write the metadata, source an image, upload it, publish. Each step was individually trivial and collectively expensive, which meant publishing cadence tracked available attention rather than editorial strategy.

The requirement was not "generate articles." It was to compress the mechanical portion of the pipeline to near zero while keeping a human editorial gate in place before anything went live.

Tool stack

Orchestration	n8n
Research	Perplexity API
Generation	GPT-4o-mini
Media	AI image → ImgBB
Publishing	WordPress REST
Control plane	Google Sheets

Architecture

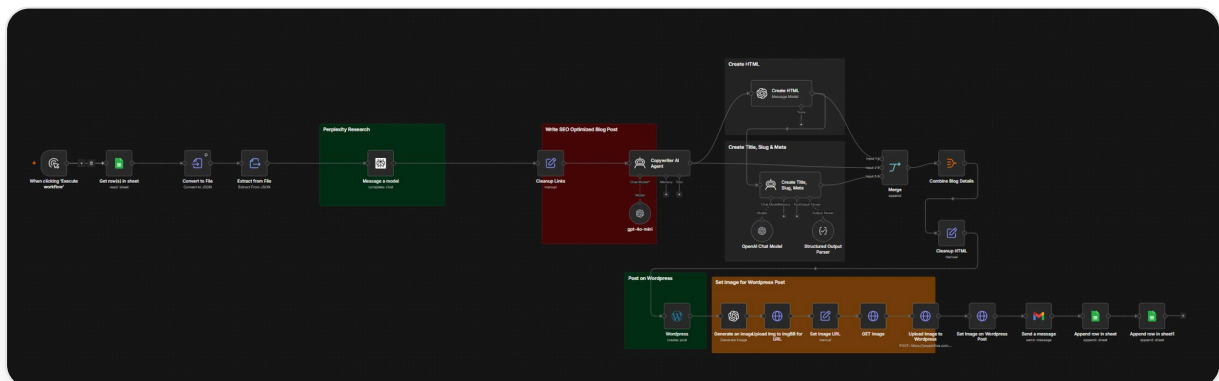
The pipeline runs two branches from a single trigger. Both converge on the same WordPress post, so the draft is never blocked on image-render latency.

Main branch

- **Google Sheets**
Trigger and topic intake
- **Perplexity**
Trend and source research
- **GPT-4o-mini agent**
Article body, HTML structure, SEO metadata
- **WordPress REST API**
Draft creation

Parallel branch

- **AI image generation**
Featured image render
- **ImgBB API**
Upload, returns a public URL
- **WordPress**
Set as featured image on the created post



SEO content engine — full pipeline view.

Technical edge

Research precedes generation

Perplexity executes before the copywriter agent, and its output is injected into the generation context. This is the difference between a model writing from parametric memory and a model writing from retrieved, current sources. It is also what makes the output defensible to an editor.

The output contract is enforced, not requested

The agent returns structured HTML plus a defined metadata object, not prose to be post-processed. Constraining the model's output shape is what makes the WordPress push deterministic — the publishing step never has to guess at parsing.

The ImgBB indirection layer

The WordPress media endpoint in this configuration will not accept a base64 payload for the featured image; it requires a resolvable public URL. ImgBB sits between generation and publication purely to resolve that mismatch. It is a small piece of the diagram and the reason the image branch works at all — the kind of constraint that only surfaces once a pipeline is actually built against a live CMS.



The image branch: generation → ImgBB → featured image assignment.

Enterprise guardrails

Human editorial gate. The pipeline terminates at draft status by design. Publication remains a deliberate human action.

Deterministic output contract. Schema-constrained generation removes an entire class of downstream parsing failures.

Retry on publish. CMS and media API calls are wrapped in retry logic; a transient failure does not orphan a half-created post.

Outcome

Topic intake to publishable draft is reduced to a single spreadsheet entry. The remaining human cost is one editorial read-through.

4 min

Average topic-to-draft latency

\$0.14

Generation cost per article

Instagram lead acquisition & CRM sync

Challenge

Creator-led outreach forced a standard trade-off: personalised messages that don't scale, or templated volume that doesn't convert. Compounding this, outbound activity that lived in an email tool but never reached the CRM produced leads that were invisible to sales — contacted, then effectively lost.

Two requirements, therefore: generate genuinely per-profile messaging at volume, and close the loop into the system of record so that every touch is attributable.

Tool stack

Orchestration

n8n

Targeting

LLM hashtag generation

Acquisition

Apify actors

Generation

Gemini

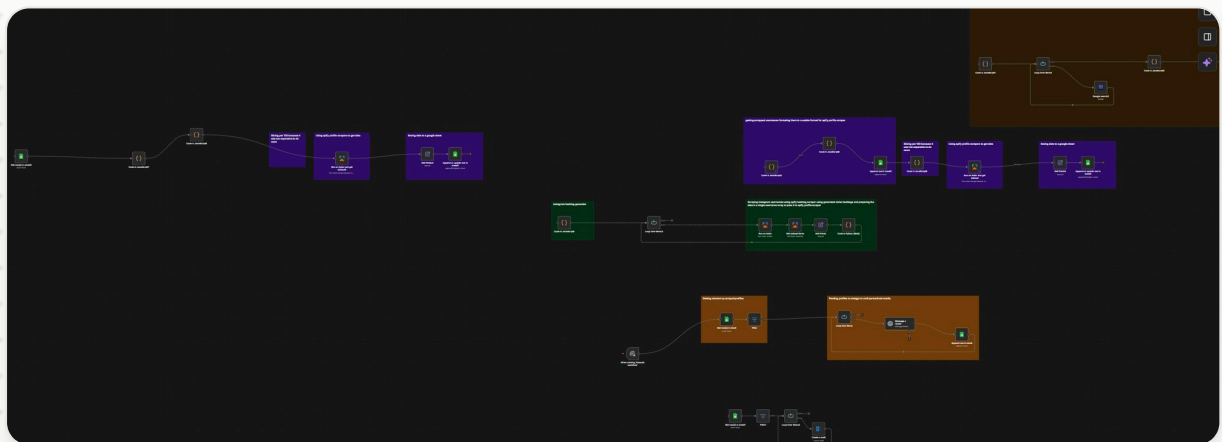
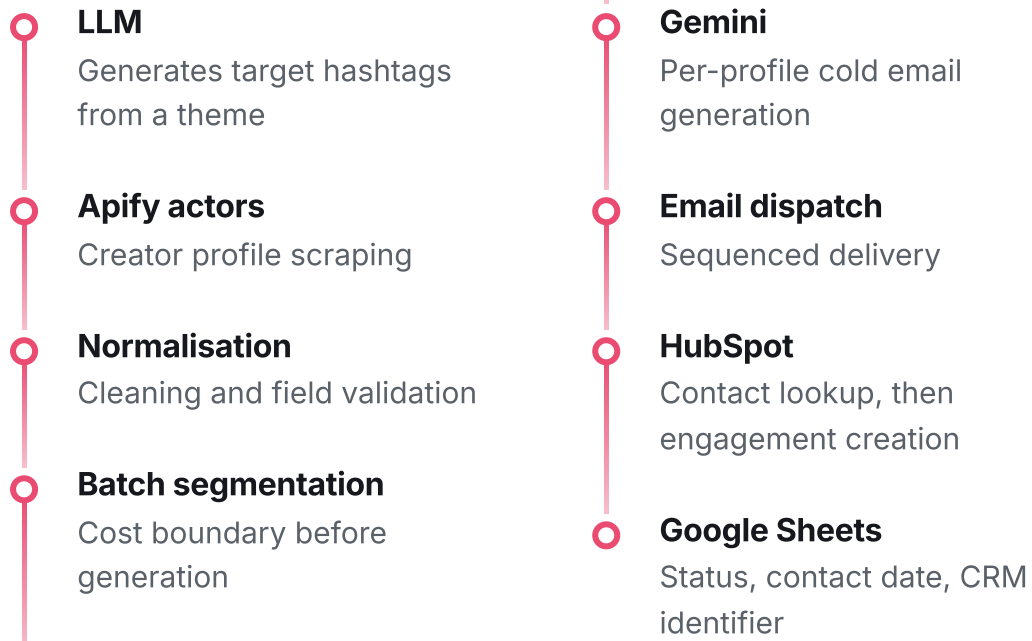
Delivery

Email sequencer

System of record

HubSpot + Sheets

Architecture



Instagram acquisition pipeline — end to end.

Technical edge

Dynamic targeting

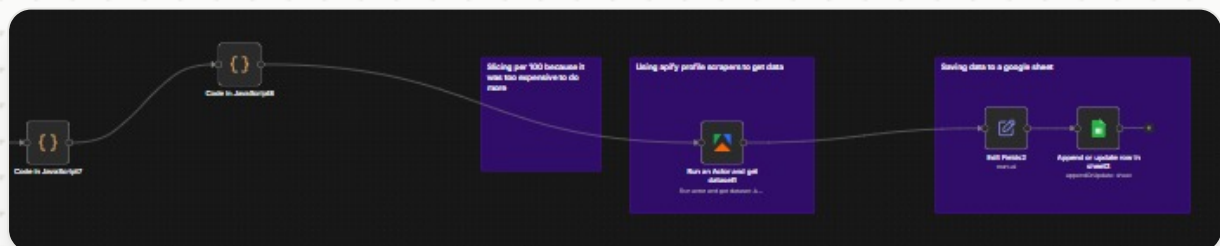
The scraping target is not a static list. A theme is passed to an LLM, which generates the hashtag set, which becomes the parameter payload for the Apify actor. The acquisition surface is redefined per run rather than maintained by hand — which means the pipeline adapts to a new vertical without a code change.

Batch segmentation as a spend boundary

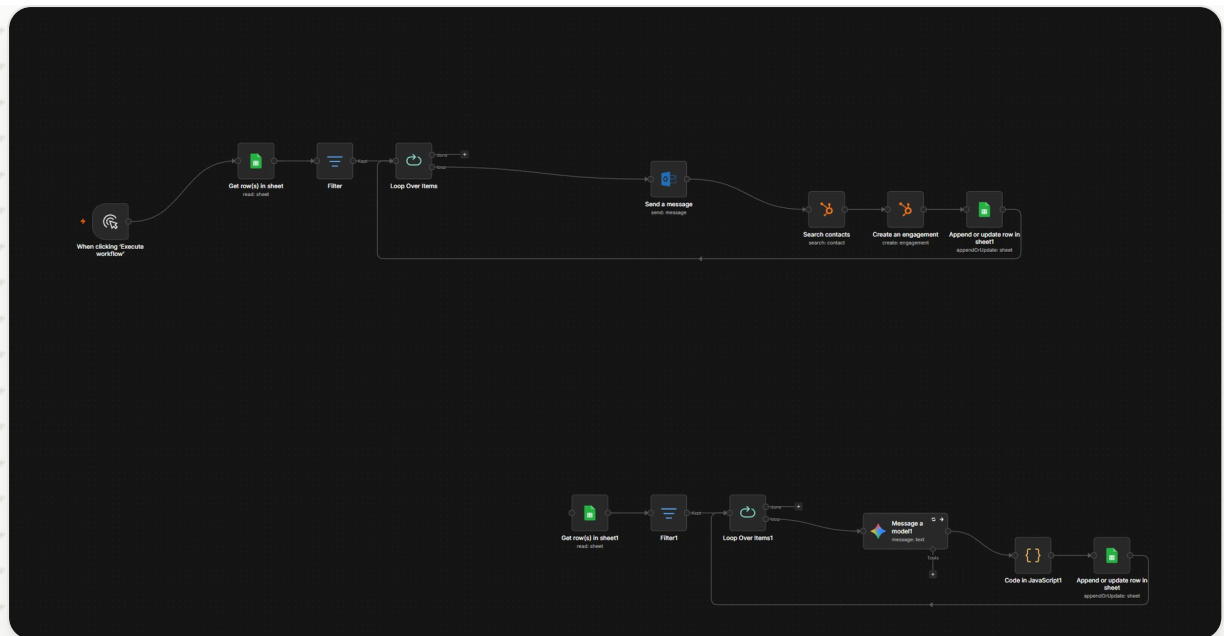
Scraped profiles are segmented into controlled batches before reaching the generation step. This is the single most consequential node in the pipeline: it converts an unbounded LLM cost into a known, per-batch figure available *before* a campaign runs. Cost per lead becomes a planning input rather than a post-hoc discovery.

Bidirectional CRM sync

The workflow does not fire and forget. After dispatch it resolves the contact in HubSpot, creates an engagement record against it, and writes the resulting CRM identifier back to the operational sheet. Sales sees full outbound history before picking up the phone; the sheet retains an operational view that reconciles with the CRM.



Batch segmentation node, configuration visible.



Dispatch, then HubSpot contact lookup, engagement creation and sheet write-back.

Enterprise guardrails

GDPR posture on acquisition. Scraped data is subject to an upstream compliance check covering source, lawful basis and traceability. An outreach pipeline without this is not deployable.

Spend ceiling by construction. Batch size is the control surface for API cost; the pipeline cannot silently exceed its budget.

Grounded generation. Message content is bounded by fields the scrape actually returned. Absent data is not inferred.

Retry with escalation. Scraping and delivery failures retry automatically and raise an alert only on persistent failure.

Outcome

Per-profile personalisation at batch volume, with every outbound touch attributable in the CRM and a known unit cost per lead.

\$0.85

Cost per qualified lead

150

Profiles processed per run

LinkedIn outreach with AI avatar video

Challenge

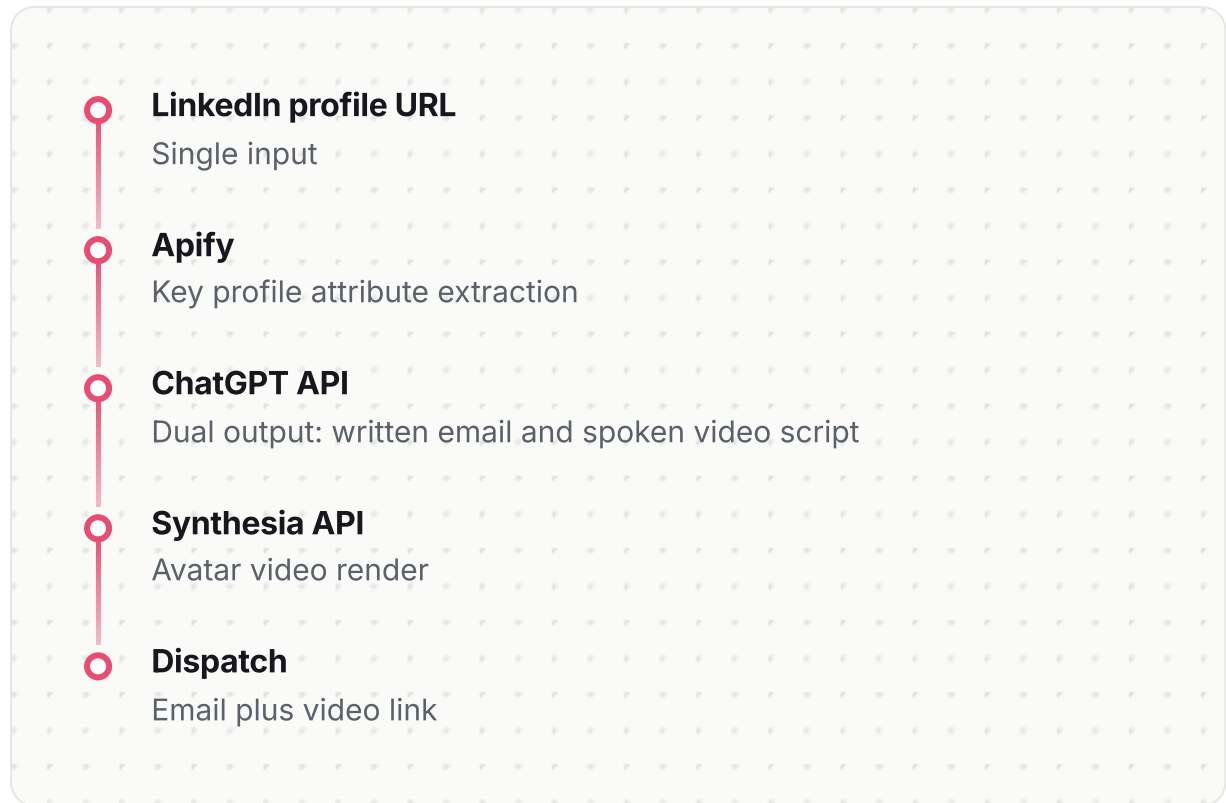
Differentiated outbound at the top of a B2B funnel is increasingly hard: written personalisation is now table stakes and commoditised. Video is materially more effective and materially more expensive, because it does not scale by construction — someone has to record it.

The objective was to produce a genuinely per-prospect video asset from a single input, without the per-asset human cost that normally makes video outreach unviable. The risk profile here is different from the other two pipelines, and it drove most of the engineering.

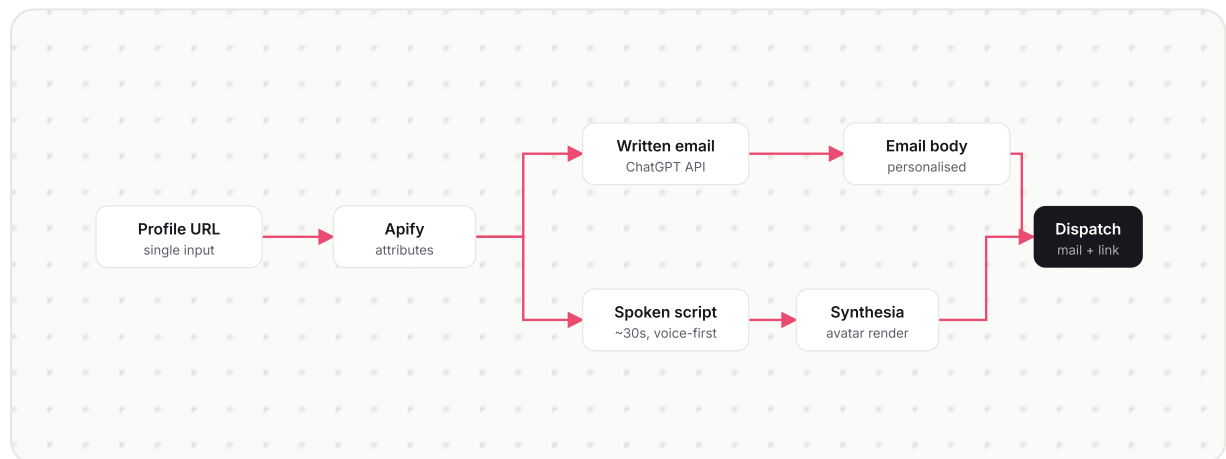
Tool stack

Orchestration	n8n
Acquisition	Apify
Generation	ChatGPT API
Video	Synthesia

Architecture



Generation fan-out



One scrape, two artefacts under different constraints, one send. The canvas for this pipeline is not archived.

Technical edge

One context, two deliverables under different constraints

The same scraped context produces both a written email and a spoken script in a single generation step. These are not the same artefact in two formats: the script must land in roughly thirty seconds when read aloud, carry no written-register jargon, and open with a hook that works by voice rather than by subject line. Both constraints are encoded in the prompt, which is where the quality of this pipeline actually lives — the Synthesia call itself is the simplest node in the diagram.

Failure mode analysis drove the design

The binding risk in this pipeline is not technical failure, it is confident error. An avatar video that states an incorrect employer or an invented job title is worse than sending nothing: the cost is reputational and it lands on a named individual. Every design decision downstream of the scrape follows from that.

Enterprise guardrails

Strict anti-hallucination constraints. The prompt prohibits the model from completing any attribute absent from the scrape. An empty field triggers rephrasing around the gap, never inference. This is the load-bearing guardrail of the entire pipeline.

GDPR verification upstream. Profile data is checked before it enters the generation path.

Token consumption monitoring. Per-execution spend is tracked, keeping video-plus-generation unit economics visible.

Automatic retry on API failure. Scraping and video-render APIs have non-trivial downtime; the pipeline absorbs it without manual re-triggering.

Outcome

A personalised video asset generated per prospect from a single URL, with generation bounded by verified upstream data.

\$2.50

Cost per generated video asset

3 min

End-to-end generation time

Reliability & operations

The three pipelines share an operational layer that is rarely visible in a demonstration and is the reason they can run unattended.

Retry and escalation

Every external API call carries retry logic. Transient failures — the common case with scraping and render services — resolve without intervention. Persistent failures escalate to an alert rather than failing silently.

Execution observability

Failed runs are inspectable down to the node that broke, with the payload that broke it. Debugging is a matter of reading state, not reconstructing it.

Cost instrumentation

Token consumption and API volume are tracked per execution across all three systems, which is what makes unit economics a known quantity rather than an estimate.

Compliance boundaries

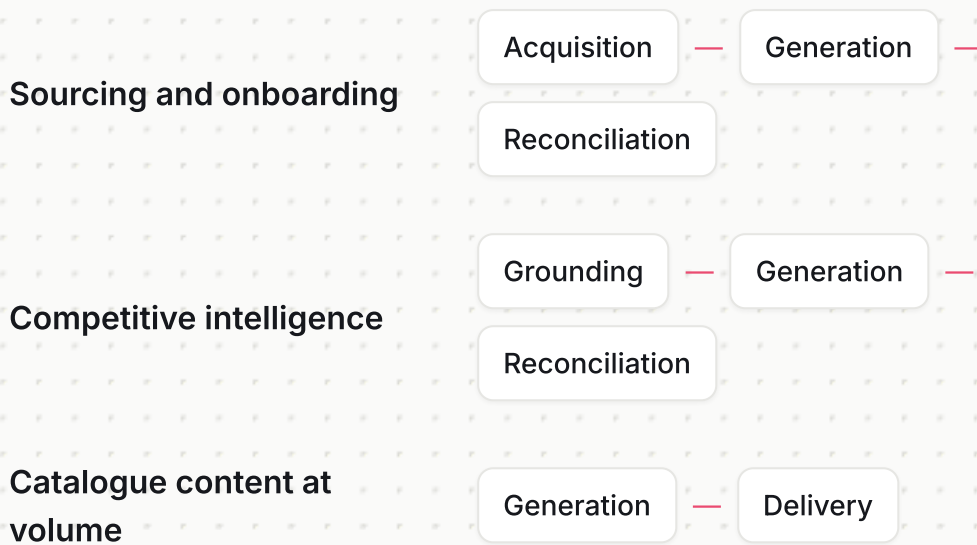
GDPR checks sit upstream of every scraping path, regardless of pipeline.

Reusable capability blocks

The three systems above are compositions of five components. They recombine to address adjacent problems without rebuilding from scratch.

Block	Function	Representative stack
Acquisition	Structured extraction from public sources	Apify actors
Grounding	Current, retrieved context for generation	Perplexity
Generation	Constrained output against an enforced schema	GPT-4o-mini, Gemini
Delivery	Push into any operational system	WordPress, email, Synthesia
Reconciliation	Write-back to the system of record	HubSpot, Google Sheets

Recombination paths



Working together

If a process in your organisation involves moving structured data between systems, generating content against it, and recording the result somewhere authoritative, it is likely addressable with the same architecture.

The starting point is usually a short scoping conversation: what the current manual path costs, what systems have APIs, and where the compliance boundaries sit.

Name

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Role

**Automation & AI pipeline
engineer**

Contact

Send me a message on Fiverr to scope out your architecture.

Production n8n Architecture — three enterprise pipelines. Prepared for client review.